

COMP4423 – Computer Vision

Assignment 1

Triangle-Brick Image Generation

[Deadline: 23:59 HKT on Tuesday, 29 September 2026]

Picture in Triangle Bricks



1. Task Description

In this assignment, you will build a triangle-brick image generation system for a real-world scenario: given an input picture, your program converts it into a combination of triangular bricks that visually approximates the original image (as in the provided example above). The generated image may use multiple brick sizes and multiple colors.

We are not overly concerned with absolute visual perfection; we care more about a complete, clearly justified pipeline, sound engineering, real-world testing, identifying problems, and finding their solutions. You may choose any algorithm and implementation approach that meets the requirements. Your report should document your step-by-step process, including your ideas, algorithm design, problems encountered, corresponding solutions, and findings.

Brick definition: Use two-dimensional **right-angled isosceles triangles**. A brick's size is the length of its two equal perpendicular sides, measured in base-grid units (e.g., sizes 1, 2, 4 and 8).

Each triangle is one brick with one uniform face color determined from its own image region. Bricks must cover the output image without gaps or overlaps. Boundaries may be drawn for display; they do not count as additional brick colors.

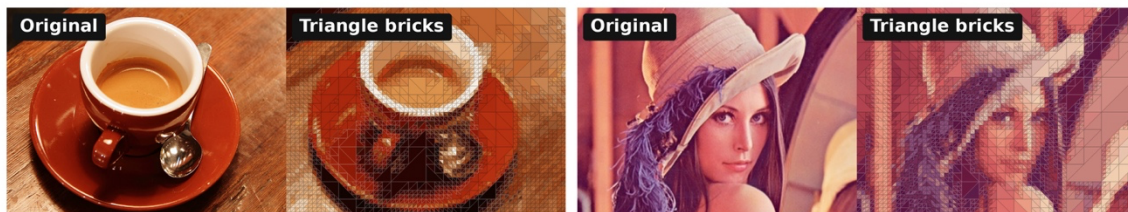
2. Tasks & Assessment

Task 1: Call the camera to **capture images and display them**. (5 marks)

Task 2*: Convert the captured image into a triangle-brick image using *equal-sized triangles* and *only three brick colors* (e.g., white, gray and black). Use **no more than 10,000 individual triangular bricks**. (10 marks)

Task 3*: Based on Task 2, use *multiple triangular brick sizes* (e.g., sizes 1, 2, 4 and 8) and *more than three brick colors*. Use no more than 10,000 individual triangular bricks. Your program should output: 1) A rendered triangle-brick image showing how bricks of different sizes are combined. 2) A brick summary: the **total number of bricks** and the **count for each brick size**. (10 marks)

Task 4*: Implement and test the program in a *real-world scenario*. Call the camera to capture images and successfully display their **triangle-brick representations** (as in the reference example below). (25 marks)



Task 5: Submit a report showing the details of your method. Questions in the report template should be answered. Explain how triangle regions, brick colors and multiple brick sizes are determined, and document your testing, problems and solutions. (50 marks)

Bonus: Submissions with excellent code quality, comment quality, output accuracy and report quality will be given a bonus of no more than 10 marks. The final grade of this assignment is $\min(100, \text{normal grade} + \text{bonus})$.

***Collaborate with Generative AI for problem-solving:** For **Tasks 2 and 3**, you are encouraged to use Generative AI to assist in developing the foundational implementation. This will demonstrate your understanding of image processing and provide you with experience in collaborating with GenAI. **Task 4** is a hands-on task where you should fully understand the implementation and code from GenAI and make necessary modifications. However, it's important to recognize that *AI can make mistakes*, so you should *identify its limitations and areas for improvement* in your report. An example prompt for Task 2 is provided at the end of this file for your reference.

3. Submission

Follow the steps below:

1. Save the report as a PDF file and rename it, *e.g.*, `Assignment1_Report_12345678d_CHAN_Dawen.pdf`
2. Compress the implementation files (*e.g.*, code) into a .zip file and rename the .zip file. *e.g.*, `Assignment1_Code_12345678d_CHAN_Dawen.zip`
3. Upload the above **two files** to the Canvas system.

Warning: If you are unable to complete the whole program, try to accomplish part of the tasks and make sure it can run successfully.

Any wrong file naming and submission will be given a ZERO mark in this assignment.

The submission deadline is **23:59 HKT on Tuesday, 29 September 2026.**

Late submission penalty: 10% is deducted for each day that the work is late. The penalty will be applied up to a maximum number of three days after and including the submission deadline day. After three days the work will be marked at zero.

This assignment is individual work. All work must be done on your own. Plagiarism is a serious offence. Copying code from web resources is prohibited as well. Any plagiarism case (for both the copier and the copiee) will be given a ZERO mark in this assignment.

4. Example Prompt for Task 2

<https://genai.polyu.edu.hk/>

The following prompt is provided as a starting point. Check and test any AI-generated code to ensure that it meets the assignment requirements.

Task description

Build a triangle-brick image generation system. Given an input picture, convert it into a combination of triangular bricks that visually approximates the original image.

Input/Output format

input: ./input.png

output: ./triangle_output.png

TODO

Convert the input image into a triangle-brick image using equal-sized right-angled isosceles triangles and only three brick colors: white, gray and black.

Determine each triangle's uniform color separately from the image region it covers. Use no more than 10,000 individual triangular bricks, counting each triangle as one brick.

The triangles should cover the output image without gaps or overlaps. Brick boundaries may be shown for display, but are not additional brick colors.

Save the generated image and explain the main processing steps.

5. Grading Criteria

Total: 100 marks

Tasks 1, 2, 3 (25 marks)

Criterion	Marks
Task 1: Capture images using a camera and display them.	5
Task 2: Use equal-sized right-angled isosceles triangular bricks.	5
Task 2: Use only three brick face colors.	5
Task 3: Use at least three triangular brick sizes and provide a brick-count summary.	5
Task 3: Use more than three brick face colors.	5

- For Tasks 2 and 3, use no more than **10,000 triangles**, with no gaps or overlaps.

Task 4 (25 marks)

Level	Criterion	Marks
1	Capture camera images and display triangle-brick results.	10
2	Additionally, test images with different content.	15
3	Additionally, demonstrate any one: real-time processing; different aspect ratios/resolutions; different environmental conditions; quantitative evaluation.	20
4	Analyze problems, implement improvements, and compare results before and after changes.	25

- Levels are cumulative; award the highest achieved score, not the sum.

Task 5: Report (50 marks)

Criterion	Marks
Problem description, motivation, and implementation details.	10
Experimental results and discussion.	10
Result analysis and reflection, including GenAI limitations where applicable.	10
Answers to all required template questions.	10
Clear organization, figures, tables, examples, and references.	10

- Deduction:** 2-5 marks for frequent spelling, grammar, or formatting errors.
- Bonus:** Up to 10 marks for exceptional code, comments, output, and report quality.
- Final assignment score capped at 100.**